

# PROBLEM SET 1: SETS

Elements, subsets, union, intersection, difference, complement, and Venn-diagram problems

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## 1. Elements of a Set

Determine whether each statement is true or false.

$A = \{2, 4, 6, 8, 10\}$

a)  $4 \in A$  b)  $7 \in A$  c)  $10 \notin A$  d)  $1 \notin A$

## 2. Write the Set

Write the set of all natural numbers less than 8.

$A = \{ \rule{1.5cm}{0.4pt} \}$

## 3. Membership

Let  $B = \{a, b, c, d, e\}$ . Insert  $\in$  or  $\notin$  in each blank.

a)  $a \rule{0.5cm}{0.4pt} B$  b)  $f \rule{0.5cm}{0.4pt} B$  c)  $d \rule{0.5cm}{0.4pt} B$  d)  $x \rule{0.5cm}{0.4pt} B$

## 4. Equality of Sets

Determine which of the following sets are equal.

$A = \{1, 2, 3, 4\}$   $B = \{4, 3, 2, 1\}$

$C = \{1, 2, 3\}$   $D = \{1, 2, 3, 4, 4\}$

## 5. Subsets

Let  $A = \{1, 2, 3, 4, 5\}$ ,  $B = \{2, 4\}$ , and  $C = \{2, 6\}$ . Decide whether each statement is true or false.

a)  $B \subseteq A$  b)  $C \subseteq A$  c)  $A \subseteq B$

## 6. Union of Sets

Let  $A = \{1, 2, 3, 4\}$  and  $B = \{3, 4, 5, 6\}$ . Find:

$A \cup B =$  \_\_\_\_\_

Work / Answer: \_\_\_\_\_

## 7. Intersection of Sets

For  $A = \{1, 2, 3, 4\}$  and  $B = \{3, 4, 5, 6\}$ , find:

$A \cap B =$  \_\_\_\_\_

Work / Answer: \_\_\_\_\_

## 8. Difference of Sets

Let  $A = \{1, 2, 3, 4, 5, 6\}$  and  $B = \{2, 4, 6\}$ . Find:

a)  $A \setminus B =$  \_\_\_\_\_ b)  $B \setminus A =$  \_\_\_\_\_

Work / Answer: \_\_\_\_\_

## 9. Union, Intersection, and Difference

Let  $A = \{2, 4, 6, 8, 10\}$  and  $B = \{1, 2, 3, 4, 5, 6\}$ . Find:

a)  $A \cup B$  b)  $A \cap B$  c)  $A \setminus B$  d)  $B \setminus A$

Work / Answer: \_\_\_\_\_

## 10. Three Sets

Let  $A = \{1, 2, 3, 4, 5\}$ ,  $B = \{3, 4, 5, 6, 7\}$ , and  $C = \{4, 5, 7, 8\}$ . Find:

a)  $A \cap B$  b)  $B \cap C$  c)  $A \cap B \cap C$  d)  $A \cup B \cup C$

Work / Answer: \_\_\_\_\_

## 11. Universal Set and Complement

Let  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$  and  $A = \{2, 4, 6, 8, 10\}$ . Find the complement of A.

$A^c = U \setminus A =$  \_\_\_\_\_

Work / Answer: \_\_\_\_\_

## 12. Sets Defined by a Rule

Let  $U = \{1, 2, 3, \dots, 12\}$ . Set A contains all even numbers, and set B contains all multiples of 3.

First write A and B. Then find: a)  $A \cup B$  b)  $A \cap B$  c)  $A \setminus B$  d)  $B \setminus A$

Work / Answer: \_\_\_\_\_

## 13. Students and Languages

In a group of 20 students, 12 study English and 9 study Russian. Five students study both languages.

Let E be the set of students who study English and R the set of students who study Russian. Find:

a)  $|E \cap R|$  b)  $|E \cup R|$  c) the number of students who study neither language.

Work / Answer: \_\_\_\_\_

## 14. Venn-Diagram Calculation

Given  $|A| = 15$ ,  $|B| = 12$ , and  $|A \cap B| = 5$ , find  $|A \cup B|$ .

Use:  $|A \cup B| = |A| + |B| - |A \cap B|$ .

Work / Answer: \_\_\_\_\_

## 15. Challenge Problem

Let  $U = \{1, 2, 3, \dots, 20\}$ . Define:

$A = \{x \in U : x \text{ is divisible by } 2\}$

$B = \{x \in U : x \text{ is divisible by } 3\}$

$C = \{x \in U : x \text{ is divisible by } 5\}$

Find: a) A b) B c) C d)  $A \cap B$  e)  $A \cup B$  f)  $A \cap B \cap C$  g)  $(A \cup B) \setminus C$

Work / Answer: \_\_\_\_\_